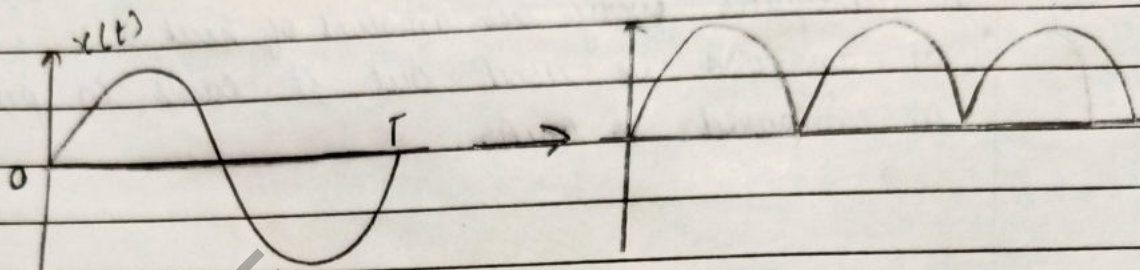


Rectifier

Rectifier $\rightarrow$ AC $\rightarrow$ Pulsating DC (AC + DC)

Unidirectional $\rightarrow$ DC

Variations $\rightarrow$ AC



$$\text{Avg value} = \lim_{T \rightarrow \infty} \frac{\int_0^T x(t) dt}{T}$$

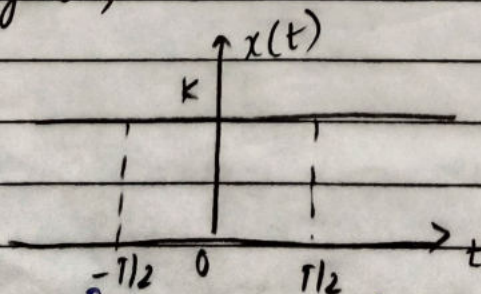
$$\text{RMS value} = \sqrt{\lim_{T \rightarrow \infty} \frac{\int_0^T x^2(t) dt}{T}}$$

For AC signal,

Avg = 0 (Area cancel outs)

Rms $\neq$ 0

For DC signal,



$$\text{Avg} = \frac{\int_0^T x(t) dt}{T} = \frac{\int_0^T K dt}{T} = \frac{K \cdot T}{T} = K$$

$$RMS = \sqrt{\frac{K^2 T}{T}} = K$$

$$Avg \neq 0, RMS \neq 0$$
$$\boxed{Avg = RMS}$$

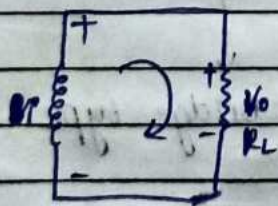
For Pulsating DC,

$$Avg \neq 0$$
$$RMS \neq 0$$
$$Avg \neq RMS$$



Half wave rectifier

Stepdown Transformer $\rightarrow$ ($\downarrow$ V for domestic use)



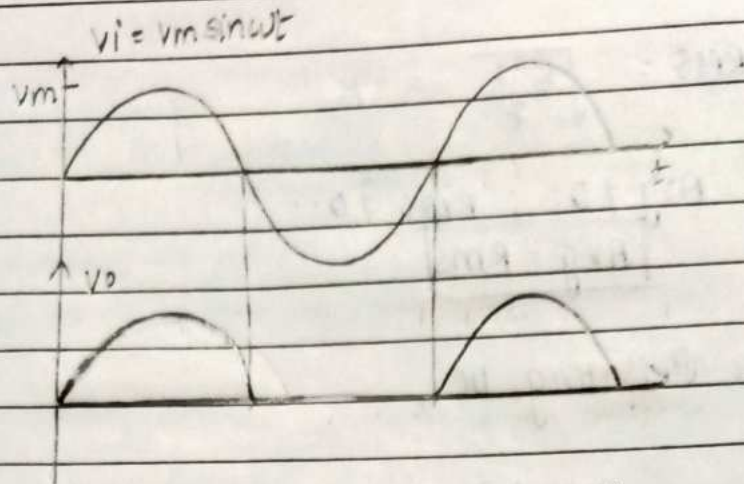
Using KVL,

$$V_i - V_o = 0$$

$$\boxed{V_i = V_o}$$

$$V_i = V_m \sin \omega t$$

$$V_o = \begin{cases} V_m \sin \omega t, & 0 < t < T/2 \\ 0, & T/2 \leq t < T \end{cases}$$

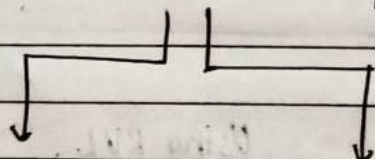


$$\eta = \frac{P_{dc}}{P_{ac}}$$

$$= \frac{\left(\frac{V_{avg}}{R_L} \right)^2 R_L}{\left(\frac{V_{rms}}{R_L} \right)^2 R_L}$$

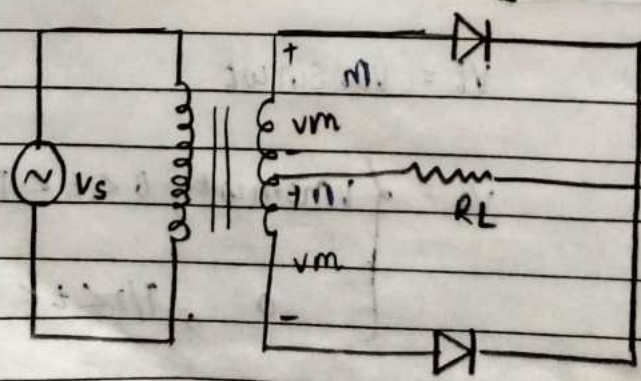
(Form factor) $FF = \frac{V_{rms}}{V_{avg}}$

Full wave rectifier

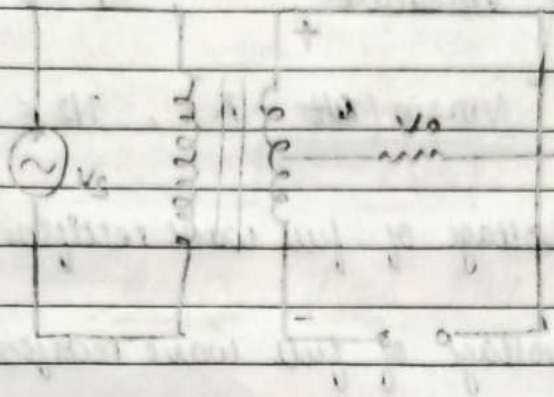


Centre tapped

Bridge type



For 1st +ve half cycle

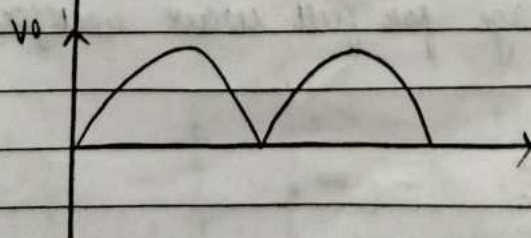
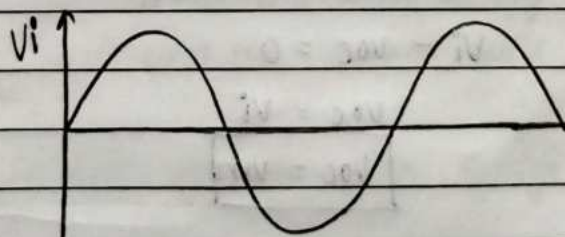
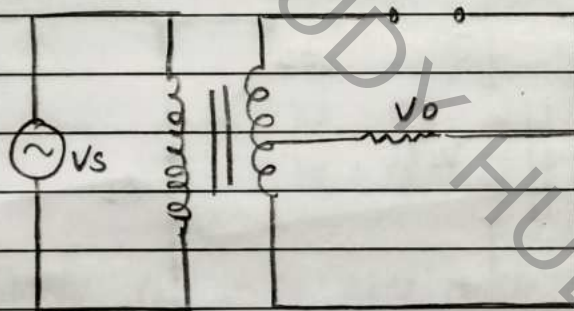


Using KVL,

$$V_i - V_o = 0$$

$$V_o = V_i$$

For -ve half cycle



$$V_o = \begin{cases} V_m \sin \omega t & , 0 < t < T/2 \\ V_m \sin(\omega t - \pi) & , T/2 < t < T \end{cases}$$

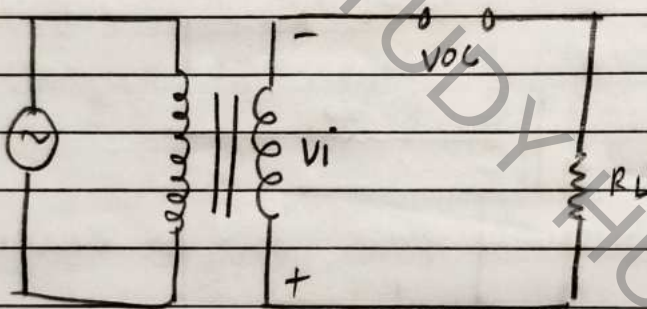
Average output voltage of full wave rectifier = $\frac{2V_m}{\pi}$

Rms output voltage of full wave rectifier = $\frac{V_m}{\sqrt{2}}$

Ripple factor ≈ 0.48

Efficiency $\approx 81.06\%$

Peak Inverse voltage of half wave rectifier

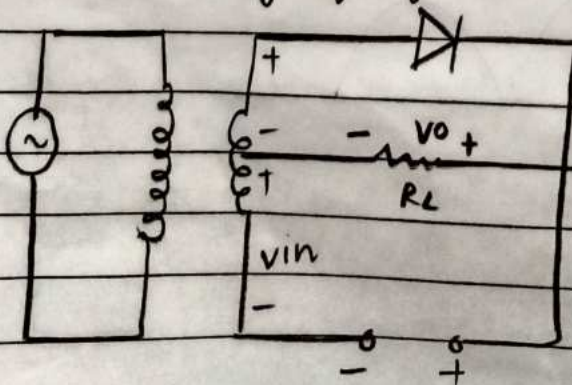


$$V_i - V_{oc} = 0$$

$$V_{oc} = V_i$$

$$V_{oc} = V_m$$

Peak Inverse voltage for full wave rectifier



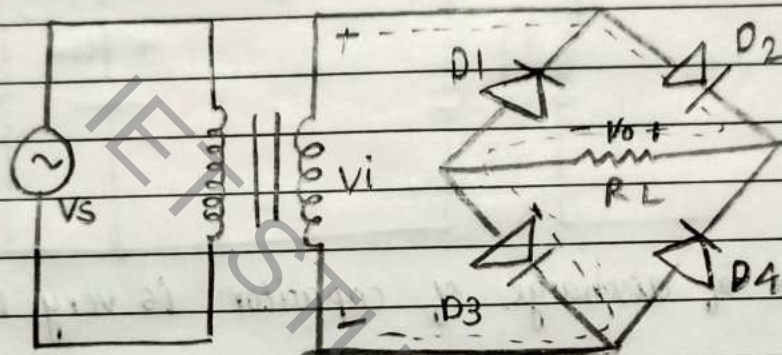
$$V_i + V_o - V_{oc} = 0$$

$$V_{oc} = V_i + V_o$$

$$= V_m + V_m$$

$$V_{oc} = 2V_m$$

Bridge Type Full wave Rectifier

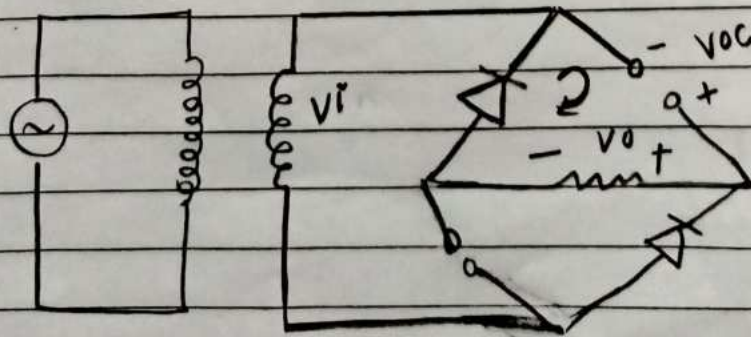


$$V_i - V_o = 0$$

$$V_o = V_i$$

Similarly for -ve half cycle,
 D_2 & D_3 will be off
 and the output will be same.

Peak inverse voltage for Bridge Type FWR

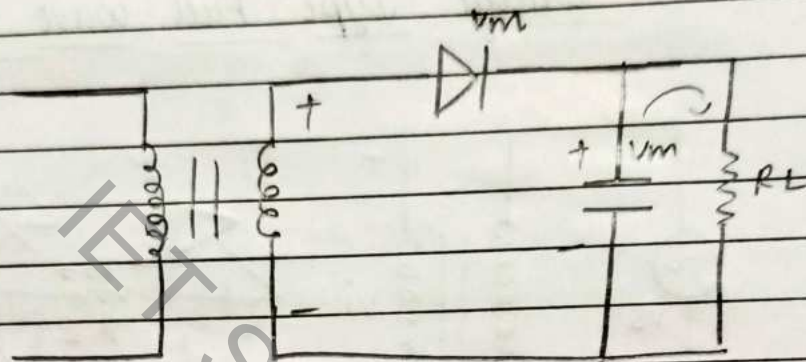


$$V_{OC} - V_O = 0$$

$$V_{OC} = V_O$$

$$V_{OC} = V_m$$

Half wave rectifier with capacitive filter.



Rate of discharge of capacitor is very low.

$$X_C \ll R_L$$

